PRD v3.0 — Week 14 Update

Property Name: Lava Birds B&B | **Island:** Hawai'i Island | **Visitor Segment:** Honeymooners

Overview of Three-Page App: The Lava Birds application consists of three main pages: a public marketing page, a public visitor dashboard, and a protected admin dashboard.

Tagline: A quiet, immersive stay near Kilauea, meant to be shared.

User (Primary Customer) Problem Statement: I am a budget-conscious traveler looking to book accommodations for my honeymoon. I can't find anything affordable online that offers unique and romantic experiences while allowing me to weigh my travel options. This is causing me stress and uncertainty.

The Opportunity

Hawai'i Island has a few existing B&B options near Kilauea, most of which offer somewhat affordable accommodations. Existing B&Bs, however, do not tailor guest experiences to reflect their proximity to Kilauea. Few offerings strongly target honeymooning couples, and packages that exist are usually found at luxury resorts. This fillable gap offers the unique opportunity to target a niche visitor segment: honeymooners seeking a romantic, immersive, and affordable experience.

Goals

The goal of the Lava Birds B&B application development is to create a responsive, easily navigable application for guests and system users. The application will aim to drive user interest, provide real-time data to help potential guests make informed decisions about travel dates, and provide an authentication-protected interface for system users.

Marketing Page

Marketing Page Requirements

Component	What is Displayed	Data Source	Interactive Behavior/ Responsive CSS Approach
Hero	Property name, tagline, island, main image/background video.	MongoDB via Express API	Hero CTA button that navigates to the booking contact form, Amenities button.
Special Offer Section	Current special offer and a CTA button prompting contact.	MongoDB via Express API	Button that opens a mailto link.
Image Carousel	Carousel of property images.	MongoDB via Express API	Navigable arrows.
About	Property description, location, target audience, lava.mp4, and owner information.	MongoDB via Express API	No major user interaction. Hover effects.
Amenities	Array of amenities: comfort-focused rooms, hot tubs, private outdoor access, local tea and coffee tastings, etc.	MongoDB via Express API	Renders amenities, no direct interaction.
CTA	Booking prompt, button, and a highlight of special offers.	MongoDB with some static fallback, as we are not developing a booking page at this stage.	Button that opens a mailto link.
Header/ Footer	Navigation links (Home, Dashboard, general nav, Admin) and copyright	Hardcoded and MongoDB/ Express API	Interactive navigation links that route users to the corresponding section.

User Personas

User Persona 1: Honeymooners

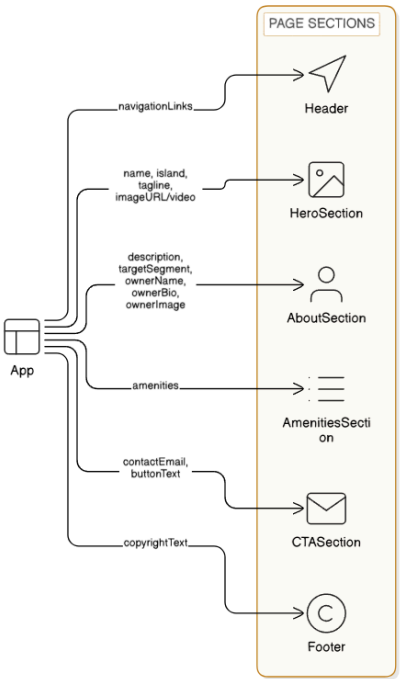
Profile/Role: A young couple who are planning a romantic honeymoon trip on a moderate budget.
Name: Maya and Ken
Goals:
- Find an affordable but memorable honeymoon experience
- Stay somewhere that feels unique and highlights the natural environment.
- Easily understand what the experience includes
- Make informed travel decisions

Pain Points:
- High costs of both unique and traditional luxury honeymoon experiences.
- Difficulty finding experiences that feel tailored to them without luxury pricing
- Potential for unpredictability in weather, volcanic activity, etc.

User Persona 2: Property Admin

Profile/Role: A non-technical property owner managing Lava Birds B&B content.
Name: Thomas
Goals:
- Easily update property information, amenities, and packages
- Post special offers and updates
- Keep the website data current without needing to update it daily
- Drive user interest

Pain Points:
- Limited technical knowledge
- Limited time to update the website daily
- Risk of outdated or inaccurate information



Original Component Hierarchy Diagram

Marketing Page Updates:

Week 13 [Weekly Edits]
Local Express server is now running with a functional fetch() call. React functional components and JSX are used throughout, with each major section added as its own component file. Updates to the database were made to tailor data and fields to Lava Birds: tagline, amenities, owner information, display images, logo, and a package offering. Alt attributes added to images comply with WCAG 2.1 AA standards. Added CORS to allow for cross-origin requests. Added a verified property ID to the code and confirmed database updates in MongoDB Compass. Designed the aesthetics and branding, ensuring a responsive layout.

Dashboard

Dashboard Design

	Arrivalchart	Originchart	Metriccards
Data Source	DBEDT visitor arrivals data	DBEDT visitor origin data	DBEDT and estimates
Chart Type	Bar Chart	Doughnut Chart	Three KPI cards
Why It Matters	Users can make informed decisions about when they'd like to arrive based on monthly trends.	Users can understand wh visits Hawai'i Island.	Show travelers market information so they can make informed travel decisions based on trends.

Island Selector Ux

UI Control Chosen State	Chosen State Variable Name	Components That Re-Render When Selection Changes	Why it Matters
Dropdown menu	slctIsland	Arrivalchart, Originchart, Metriccards, WeatherWidget	Allows users to navigate to information on their chosen Island. Must be easy to use, and components must consistently reflect the selection.

Widgets

	Placement On Page	City Queried	Loading State Handling; Data Fields Displayed	Why it Matters
Weather	Top left of the page, unde the top user interactive section	Hilo, Hawai‘i	“Loading” message, error handling; city name, temperature, conditions	Allows users to reflect on weather conditions. This can allow users to track daily weather through the site, boosting interaction.
Volcano	Top middle of the page, to the right of the weather widget	Hilo, Hawai‘i (Kilauea)	“Loading” message, error handling; eruption data, conditions	Allows users to reflect on volcanic activity. This can allow users to track activity through the site and make decisions on travel based on activity. The video may increase conversion.
Live Stream	Top right of the page, To the right of the volc. widget	Hilo, Hawai‘i (Kilauea)	“Loading” message. Potential to add fallback to another stream.	Will drive interest for users. Users will get to see a real time stream of Kilauea.

Data Architecture

DBEDT Fields Used	JSON Schema Sketch	Express Route That Serves The Data
Island, Month, Arrivals, Metrics	{“ island”: "Hawai'i", “month”: "Feb", “arrivals”: 182000, “market” (by mo): {“japan”: 20000, “canada”: 8000, “other”: 9000 }, “metrics”: {“lengthofstay”: 8.9, “expenditure”: 250, “occupancy”: 73%}}	GET /api/arrivals?island=Hawai’i

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Authentication & Security

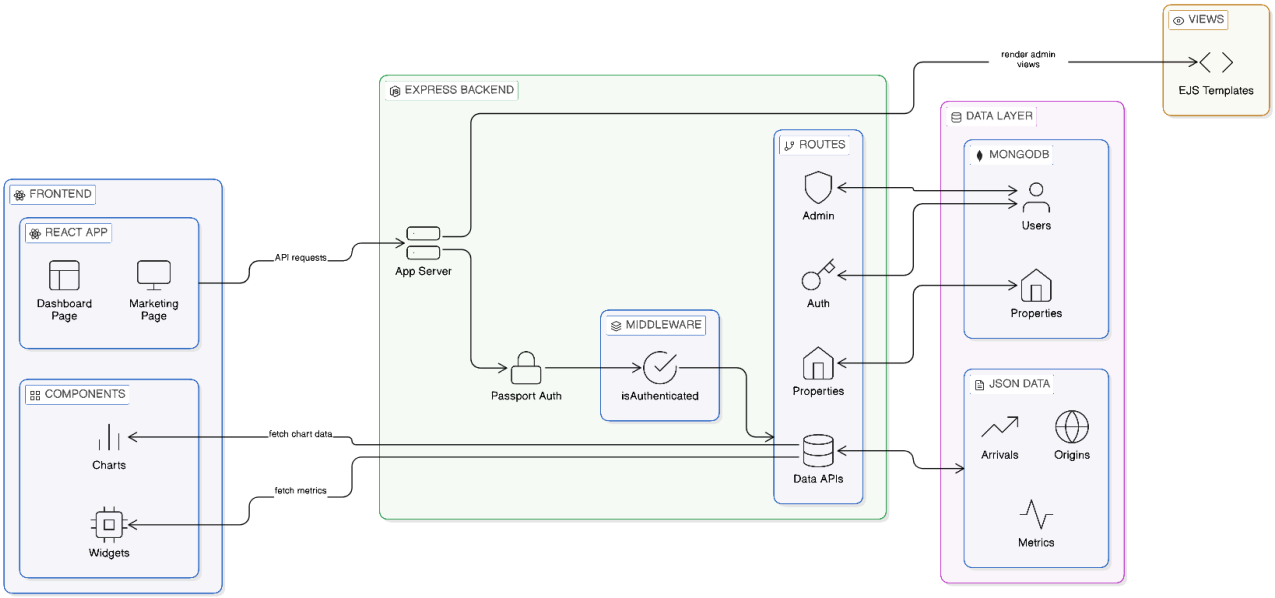
Authentication Requirements

- Authentication Method- Verifying Who the User Is: Email + password via Passport.js LocalStrategy.
- Admin User: An admin user has elevated privileges. They have the ability to make changes and view data that a normal system user cannot. Lava Birds B&B admins will be individuals granted these privileges, i.e., managers, specified employees, owner, etc.
    - After authentication, Lava Birds B&B Admins can:
        - Access a table of property documents from MongoDB
        - Display the count of guest reviews across all properties
        - Post announcements and special offers on the marketing page
- Protected Routes:
    - Admin/Login: All users can access the admin login page. Not all users can proceed past this page. This is the page admin users will use to log in.
    - /Admin/Dashboard: protected admin-only route. This route is used to access property information.
    - /Admin/Accounements: protected admin-only route. Used to post announcements to the marketing page.
- Login Flow:
    - **Prior req:** admin registers via seed-admin.js
    - Admin visits “Admin” page
    - Admin enters email/ password
    - Form submits
    - Passport strategy checks user credentials on MongoDB
    - bcrypt checks the entered password against the stored hash (with a minimum of 10 salt rounds)
        - If validated, the user redirects to /admin/dashboard
        - If invalid, the user is redirected back to the login page, and an error message is displayed.
- Session Management:
    - Session Time: The session will automatically end after one hour for security purposes. Admins must then log back in.
    - When an admin closes the browser, they will be automatically logged out (may depend on cookie settings--pending development and testing).

Security Requirements

- Password Hashing:
  - Passwords are hashed using bcrypt with a minimum of 10 or more salt rounds
  - Storing passwords as plain text will NOT be used. This is important as if passwords are stored as plain text, anyone with database access can see all passwords. This is a major security risk. For example, if someone hacked into the database, they would not need to take any other measures to access passwords. They would be unencrypted and displayed as plain text.
- Environment Variable Management:
  - SESSION_SECRET and MONGODB_URI, and API keys will be in .env. The .env example will be committed to GitHub to provide an example of a functional .env with placeholders for sensitive keys and information. .env will be listed in .gitignore to ensure these are not posted publicly.
- Handling Failed Login Attempts:
  - When a user inputs an incorrect password, they will be:
    - Redirected to /admin/login
    - An error message stating “Invalid credentials. Please try again.” will display.
- Additional Security Risk to Consider:
  - XSS in review forms:
    - XSS attacks occur when attackers inject malicious scripts into trusted websites. Then, when users visit the page with the affected scripts, the browser executes this code, causing multiple security issues. This is an important consideration for Lava Birds B&B, as unprotected review forums could potentially allow for this type of attack.
    - Potential Protections:
      - Filter input on arrival
      - Limit form fields to expected values
        - Use appropriate response headers
- **Significant Authentication Design Decisions:** The decision to hash passwords with bcrypt is the most significant design choice, as it will protect from attacks. If passwords were stored as plaintext, this would be a very high security risk. The choice to potentially implement session time-out measures is another significant security choice. This can prevent unlimited access sessions.

Updated Architecture Diagram



AI Disclaimer: AI design tools (Eraser.io) were used to create the component architecture diagram.

User Stories

User Persona 1: Honeymooners

Profile/Role: A young couple who are planning a romantic honeymoon trip on a moderate budget.
Name: Maya and Ken

- “As a user and guest, it is very important to me that my information is kept private so that I can feel safe inputting my information and booking with Lava Birds B&B. It’s also very important to me that my stored information is not altered in any way by unauthorized users. I’d also like announcements to only be posed by authorized personnel”.

User Persona 2: Property Admin

Profile/Role: A non-technical property owner managing Lava Birds B&B content.
Name: Thomas

- “As a system administrator and manager, it is very important to me that only authenticated users can access guest information and post anything to the site. If an attacker were able to get in and post an unauthorized announcement with a phishing link for users, this would be a nightmare. It’s also very important to me that my password is kept safe”.

Considerations for Weeks 14 And 15

Original Week 13 Authentication Integration Plan

Implement admin authentication using Passport.js (local strategy) or JWT and create an admin login page. Store hashed passwords in MongoDB using bcrypt, implement CSRF protection, and add an admin-only feature allowing postings of special offers or announcements.

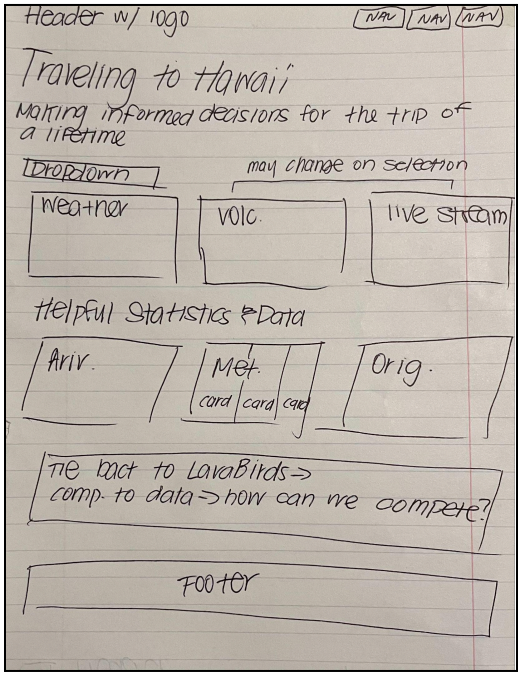
Week 13 Open Design Questions

How will I refine the general aesthetic design? Do I still need to transition any hardcoded data to MongoDB? How will I implement authentication protocols? Where will announcements be posted, in what design format? How do I design the island selector? What additional islands should be included in the weather widget? Is the site easy to navigate? Does the site resolve both the problem statement and user pain points? What is the predicted conversion rate?

Important Acceptance Criteria for All Pages

Marketing Page:

- **Given** that a user visits the marketing page, **when** the page loads, **then** the lava video plays automatically without substantial load time and loops without interruption.
- **Given** that a user visits the marketing page, **when** they navigate around the page, **then** they book a stay at a 4% conversion rate.
- **Given** that a user visits the marketing page, **when** they navigate around the page, **then** they are able to easily identify amenities and booking processes within 20 seconds without confusion.



Dashboard Wireframe First-Draft

Dashboard:

- **Given** that a user visits the dashboard page, **when** the weather widget displays, **then** it loads within 1-2 seconds and displays the loading message while loading.
- **Given** that a user changes their island selection, **when** their choice is input, **then** Arrivalchart, Originchart, Metriccards, and WeatherWidget update within 1-2 seconds.
- **Given** that a user visits the dashboard page, **when** they would like to navigate to a specific section, **then** they are able to quickly and easily identify how to do so within 20 seconds without frustration, getting meaningful information from their search.

Admin:

- **Given** that a user visits the admin page, **when** they enter an invalid password, **then** they will be redirected to the login page, and the error message will display.
- **Given** that an admin visits the admin page, **when** they enter their password and are successfully authenticated, **then** they are able to access information that helps them understand property data in an efficient way.
- **Given** that an admin visits the admin page, **when** they enter their password and are successfully authenticated, **then** when they post an announcement, it successfully posts to the marketing page.